10TH CLASS PHYSICS — NEW SYLLABUS (PECTAA)

Chapter 16 — Electricity

Solved Exercise | New Book (PECTAA 2026)

Exercise 16 (Solved)

This exercise covers electric current, e.m.f. and potential difference, Ohm's law, resistivity, series and parallel circuits, electrical power and energy, and the hazards of electricity with safety measures.

A. Multiple Choice Questions

16.1 The direction of conventional current is:

(a) from negative to positive terminal
(b) from positive to negative terminal
(c) in both directions at the same time
(d) it does not have a fixed direction

Answer: (b) from positive to negative terminal — conventional current is taken in the direction in which positive charge would flow, i.e. out of the positive terminal, through the circuit, to the negative terminal.

16.2 What type of electric current flows in one direction only?

(a) Alternating current
(b) Direct current
(c) Induced current
(d) None of these

Answer: (b) Direct current — in direct current (d.c.) the electrons flow steadily in one direction, as from a cell or a battery; alternating current reverses its direction many times a second.

16.3 The device used to measure electric current:

(a) voltmeter
(b) ammeter
(c) ohmmeter
(d) wattmeter

Answer: (b) ammeter — an ammeter measures current in amperes and is always connected in series with the component.

16.4 What is the function of an electromotive force (e.m.f.) source in a circuit?

(a) To reduce resistance in the circuit
(b) To provide energy to drive charges through the circuit
(c) To store electrical energy for later use
(d) To convert electrical energy into mechanical energy

Answer: (b) To provide energy to drive charges through the circuit — the source (cell, battery or generator) supplies the energy per unit charge that keeps the current flowing round the circuit.

16.5 The relationship between voltage, current, and resistance is:

(a) $V = I\,R$
(b) $V = I / R$
(c) $V = I^2\,R$
(d) $V = R / I$

Answer: (a) V = I R — Ohm's law: the potential difference across a conductor is directly proportional to the current through it, the constant being the resistance.

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16.6 What happens to the resistance of a metal conductor when its temperature increases?

(a) It decreases
(b) It remains constant
(c) It increases
(d) It becomes zero

Answer: (c) It increases — on heating, the atoms of the metal vibrate more strongly and obstruct the flow of electrons, so the resistance of a metallic conductor increases.

B. Short Answer Questions

16.1 How does an ammeter measure electric current?

An ammeter is a moving-coil or digital instrument that measures the rate of flow of charge through a circuit. It is always connected in series with the component whose current is to be measured, so that the whole current of the circuit passes through it. Its reading is in amperes.

It works on the magnetic effect of current: the current passing through the coil of the ammeter produces a magnetic field which deflects the coil against a spring, and the deflection is proportional to the current. An ideal ammeter has a very low resistance, so that it does not change the current it is measuring.

16.2 What is the main difference between direct current and alternating current?

Point	Direct current (d.c.)	Alternating current (a.c.)
Direction of flow	flows in one direction only	reverses its direction again and again
Magnitude	stays constant with time	changes continuously, usually as a sine wave
Source	cells, batteries, solar panels, d.c. generators	a.c. generators (alternators), mains supply
Frequency	zero (no frequency)	50 Hz in Pakistan (changes direction 100 times per second)
Transmission	not suitable for long distances, large losses	easily stepped up or down by a transformer, so efficient for transmission
Uses	electronic devices, torches, phones, cars	home wiring, motors, fans, industrial machinery

16.3 Define electromotive force (e.m.f.).

The electromotive force of a source is the energy supplied by the source to move a unit positive charge once completely round the circuit. It is equal to the potential difference across the terminals of the source when no current is being drawn, i.e. on open circuit.

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In symbols, e.m.f. E = W / q, where W is the work done by the source in carrying the charge q round the circuit. Its unit is the volt (1 V = 1 J C^{-1}). Although the name contains the word 'force', e.m.f. is not a force but a quantity of energy per unit charge.

16.4 State Ohm's law and define resistivity.

Ohm's law: the current flowing through a metallic conductor is directly proportional to the potential difference across its ends, provided that the temperature and other physical conditions remain constant. So V ∝ I, or V = I R, where R is the resistance of the conductor.

Resistivity is the resistance of a conductor of unit length and unit area of cross-section; it is a property of the material itself and does not depend on the size or shape of the conductor. It is given by ρ = R A / l and its unit is the ohm metre (Ω m).

The resistance of a wire depends on its length, its area of cross-section and the material, by R = ρ l / A. Copper has a resistivity of about 1.7 × 10^{-8} Ω m and is used for wires, while nichrome, about 1.1 × 10^{-6} Ω m, is used for heater elements.

16.5 What is the formula for calculating the equivalent resistance of resistors connected in series?

When resistors are joined in series, the same current passes through each of them and the total potential difference is shared between them. Therefore the equivalent resistance is the sum of the individual resistances:

$$R_s = R_1 + R_2 + R_3 + \ldots$$

For example, for 4 Ω, 6 Ω and 8 Ω in series, R_s = 4 + 6 + 8 = 18 Ω. The equivalent resistance in series is always greater than the largest of the individual resistances.

16.6 How does temperature affect the resistance of a conductor?

The resistance of a metallic conductor increases when its temperature rises. On heating, the atoms and ions of the metal vibrate with a larger amplitude about their positions and obstruct the flow of electrons more strongly; the number of collisions increases, so the resistance increases. For a metal, the resistance at temperature T is $R = R_0(1 + \alpha\Delta T)$, where α is the temperature coefficient of resistance.

In semiconductors such as germanium and silicon the opposite happens: as the temperature rises, many more free charge carriers become available, so the resistance decreases. That is why a thermistor (a semiconductor) is used to measure and respond to temperature. Carbon also behaves like a semiconductor, while alloys such as manganin change very little with temperature and are used for standard resistors.

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C. Constructed Response Questions

16.1 Why is it important that an ammeter has very low resistance and is connected in series, while a voltmeter has very high resistance and is connected in parallel? What would happen if they were connected incorrectly?

An ammeter is used to measure the current through a component, so it must be connected in series with that component and all the current has to pass through it. If it had a large resistance, it would add to the resistance of the circuit and reduce the current, so the reading would not be the true current. Therefore an ideal ammeter has a very small resistance, almost zero, so that it disturbs the circuit as little as possible.

A voltmeter is used to measure the potential difference across a component, so it is connected in parallel with that component. If it had a low resistance, it would form an easy path across the component and a large current would flow through the meter itself, saving little or no current for the component and changing the whole circuit. Therefore a voltmeter has a very high resistance, so that almost no current passes through it.

If they are connected wrongly: an ammeter joined in parallel would offer a nearly zero-resistance path across the supply, producing a very large current which would burn out the meter and blow the fuse — a short circuit. A voltmeter joined in series would put a huge resistance in the circuit, so the current would fall almost to zero and the device would not work; the meter would read nearly the full supply voltage. This is why the two meters have quite different resistances and are always connected the way they are.

16.2 If a copper wire is replaced with an aluminum wire of the same dimensions, how would the current and power consumption change in the same circuit? Explain using the concept of resistivity.

Resistance is given by $R = \rho\, l / A$. With the same length and the same area of cross-section, the resistance depends only on the resistivity ρ. The resistivity of aluminium (about 2.6×10^{-8} Ω m) is greater than that of copper (about 1.7×10^{-8} Ω m), roughly 1.6 times as large.

So the aluminium wire has a larger resistance, about 1.6 times that of the copper wire. Since the same supply voltage is applied, the current is $I = V / R$, so the current becomes smaller — about 0.6 times the earlier value. The wire will carry less current and the lamp or appliance will get less current too, so it will work below its proper power.

Power consumption in the wire itself is $P = I^2R = V^2 / R$. Because R is larger and I is smaller, the heat produced per unit length of the connecting wire increases ($P_{wire} = I^2R$), so the aluminium wire becomes warmer and there is more energy loss as heat. So replacing copper with aluminium of the same size reduces the current, makes the appliance less effective and loses more energy in the wires — that is why aluminium cables must be made thicker than copper ones for the same current.

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16.3 In a household lighting circuit, what practical problems might arise if the bulbs were connected in series instead of parallel?

1. All bulbs would carry the same current and the supply voltage would be shared among them. With more bulbs the share of each is less, so every lamp would be dim, and the brightness would change whenever a bulb was switched on or off.
2. If one bulb failed (its filament broke), the circuit would be broken and all the lamps would go out, because there is only one path for the current. In parallel connection, a failed lamp does not affect the others.
3. Each bulb could not be switched on or off independently; a single switch in the series loop would control the whole circuit.
4. The total resistance would increase with every lamp added ($R_s = R_1 + R_2 + \ldots$), so the current would keep falling and the lamps would grow dimmer, and the wiring would have to be designed for the whole series chain.
5. Appliances of different power ratings cannot work correctly, because the voltage across each depends on the resistance of the others. For these reasons household circuits are connected in parallel, where each lamp gets the full 220–230 V, works independently and can be switched separately.

16.4 You are tasked with designing a fuse for a device that operates at 230 V and 5 A. What rating would you choose for the fuse and why? What could go wrong if the rating is too low or too high?

The device draws a normal current of 5 A, so the fuse must carry this current without melting. A fuse is chosen for a current slightly above the normal working current, so a 6 A fuse (or a 5 A fuse of good quality, though 6 A is safer) is the right choice. The fuse rating must never be below the working current, otherwise it would blow in normal use.

If the rating is too low: the fuse will melt and cut off the supply even when the device is working normally, so the device keeps switching off for no fault — 'nuisance blowing'. Repeated blowing can also make people replace it with a thicker wire, which is dangerous.

If the rating is too high: the fuse will not melt when a fault occurs. An overload or short circuit will then overheat the wiring, the plug or the device itself, and can start a fire or damage the appliance, before the fuse ever acts. So the fuse must be matched to the current the circuit is designed to carry — not too low and not too high.

16.5 Explain why A.C. is more suitable for power transmission across cities, even though D.C. is used in batteries and portable electronic devices.

The great advantage of alternating current is that its voltage can be changed easily and efficiently with a transformer. A.C. can be stepped up to a very high voltage at the power station and stepped down again near the consumers, using transformers that have no moving parts and very little loss. Direct current cannot be transformed in this way.

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High-voltage transmission is important because the power lost in the cables is $P_{loss} = I^2R$. For a given power, P = VI, raising the voltage lowers the current, and the loss in the line falls as the square of the current. So the same power can be sent across a city or a country through thin cables with very small losses. With d.c. at a moderate voltage the current would have to be huge, the cables very thick and the losses enormous.

A.C. generators are also simpler and cheaper than d.c. generators, and a.c. motors are robust and need little maintenance, so a.c. is used for homes, fans and industry. D.C., on the other hand, is needed in batteries and portable devices because a cell always gives current in one direction, and electronic circuits such as chips, LEDs and phone chargers need a steady d.c. supply. So a.c. is used for the generation and transmission of power, while d.c. is used inside electronic devices.

D. Comprehensive Questions

16.1 Explain the concept of electric current. How is it related to the movement of charges, and how is it measured in a circuit?

Electric current is the rate of flow of electric charge through a conductor. If a charge Q passes through a cross-section of the conductor in time t, the current is I = Q / t. Its unit is the ampere: 1 A = 1 C s^{-1}, that is, one coulomb of charge passing a point every second.

Relation to the movement of charges. In a metal the free electrons are the charge carriers. When a potential difference is applied across the ends of the wire, an electric field is set up inside it and the free electrons drift towards the positive terminal, opposite to the field. Their drift is slow (about a millimetre per second) but the electric effect travels almost at the speed of light, so a lamp lights at once. By convention the direction of current is taken as the direction in which positive charge would move, i.e. opposite to the electron flow. In electrolytes both positive and negative ions move; in gases and semiconductors the carriers are ions and electrons or holes.

How it is measured. Current is measured with an ammeter, connected in series with the component so that the whole current passes through it. A galvanometer with a shunt, or a digital meter, is used; the ammeter has a very low resistance so that it does not change the current. The current can also be calculated from Ohm's law, I = V / R, if the voltage and the resistance are known, and small currents are measured in milliamperes or microamperes.

16.2 Define electromotive force (e.m.f.) and potential difference. How does a battery create a potential difference, and what is the role of e.m.f. in maintaining the flow of current?

Electromotive force of a source is the energy supplied by the source to carry a unit positive charge once round the whole circuit: E = W / q, measured in volts.

Potential difference between two points is the work done (energy converted) when a unit positive charge moves from one point to the other through the external circuit: V = W / q, also in volts. In practice, the e.m.f. is the value across the terminals when

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no current flows (open circuit), and the potential difference is the value across the terminals while the current flows; it is smaller than the e.m.f. because some energy is used up in overcoming the internal resistance r, so V = E − I r.

How a battery creates a potential difference. In a cell, chemical reactions at the two electrodes push electrons to the negative terminal and remove them from the positive one. This builds up a surplus of negative charge on one terminal and a deficit on the other, so a potential difference appears between them. The chemical energy of the cell is thus converted into electrical energy, and the cell acts as a pump for charge.

Role of the e.m.f. When the terminals are connected by a wire, the potential difference drives a current through it. As the charge flows, the terminals would quickly become discharged, but the chemical reactions inside the cell keep transferring charge and maintaining the potential difference. The e.m.f. is thus the energy per unit charge that the source keeps supplying to maintain the flow; without it, the current would stop at once. A larger e.m.f. means more energy per coulomb and hence a larger current for the same resistance.

16.3 State and explain Ohm's law. How does the resistance of a conductor affect the flow of current, and how can Ohm's law be applied in practical electrical circuits?

Ohm's law: the current through a metallic conductor is directly proportional to the potential difference across its ends, provided the temperature and other physical conditions do not change. So I ∝ V, or V = I R, where the constant R is the resistance of the conductor.

Explanation and limits. The law holds for metallic conductors (ohmic conductors), and a graph of V against I is a straight line through the origin whose slope is R. It fails for non-ohmic devices such as filament lamps, diodes and thermistors, because their resistance changes with temperature or voltage. Resistance is the opposition offered by the conductor to the flow of charge; it is measured in ohms, and 1 Ω = 1 V A^{-1}.

Effect of resistance. For a fixed voltage, the current is inversely proportional to the resistance (I = V / R): doubling the resistance halves the current. Resistance itself depends on the material (resistivity), on the length (longer wire, more resistance) and on the area of cross-section (thicker wire, less resistance), by R = ρ l / A; it also increases with temperature for a metal.

Practical applications: choosing a resistor in series with an LED to limit the current; using a potential divider of two resistors to obtain a smaller voltage from a supply; the resistance wire of a heater and the filament of a lamp; calculating the current drawn by an appliance from its rated voltage and resistance; and finding the unknown resistance of a wire by measuring V and I in the laboratory experiment. Ohm's law is thus the basic tool of circuit calculation.

16.4 Differentiate between series and parallel combinations of resistors. Provide mathematical expressions for calculating the total resistance in both types of circuits.

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Point	Series combination	Parallel combination
Connection	resistors joined end to end, one path only	all left ends joined to one point and all right ends to the other, several paths
Current	same current through every resistor	current divides; different current in each branch
Potential difference	supply voltage divides among the resistors	same voltage across every branch
Total resistance	$R_s = R_1 + R_2 + R_3 + \dots$	$1/R_p = 1/R_1 + 1/R_2 + 1/R_3 + \dots$
Value of total R	greater than the largest resistance	smaller than the smallest resistance
If one component fails	the whole circuit is broken and stops working	only that branch stops; the rest keep working
Uses	decorative light strings, fuses, adding cell voltages	household wiring, car headlights, voltmeter connections

Examples: for 4 Ω, 6 Ω and 8 Ω in series, $R_s = 18$ Ω. For 10 Ω and 30 Ω in parallel, $1/R_p = 1/10 + 1/30 = 4/30$, so $R_p = 7.5$ Ω. The parallel value is always smaller than either resistor, because the extra path makes it easier for the current to flow.

16.5 What is meant by resistivity of a material? How does it depend on the material, and what role does temperature play in changing the resistivity of conductors and semiconductors?

Resistivity is the resistance offered by a conductor of unit length and unit cross-sectional area made of that material. From $R = \rho l / A$, the resistivity is $\rho = R A / l$, and its unit is the ohm metre (Ω m). Resistivity is a characteristic property of the material and does not depend on the size or shape of the sample, so it is used to compare materials.

Dependence on the material. Metals have low resistivity because they have many free electrons — copper about 1.7×10^{-8} Ω m, silver even lower. Alloys such as nichrome have much higher resistivity (about 1×10^{-6} Ω m) and are used in heater elements because they get hot without melting. Insulators such as rubber and glass have resistivities of the order of 10^{12} to 10^{16} Ω m, and semiconductors such as germanium and silicon lie in between, about 10^{-1} to 10^{3} Ω m.

Role of temperature. In a metal, raising the temperature makes the ions vibrate more vigorously, so the electrons collide more often and the resistivity increases — nearly linearly, $\rho = \rho_0(1 + \alpha\Delta T)$. In a semiconductor, raising the temperature breaks many more bonds and releases many more electron–hole pairs; the number of charge carriers rises so quickly that the resistivity decreases strongly, even though the ions vibrate more. This is why a thermistor's resistance falls as it is heated, and

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why semiconductors are used in temperature sensors and thermometers.

E. Numerical Problems

16.1 A charge of 25 coulombs passes through a circuit in 10 seconds. Calculate the current flowing through the circuit.

Given: Q = 25 C
t = 10 s

To find: I = ?

Formula: I = Q / t

Solution: I = 25 / 10
I = 2.5 A

Answer: I = 2.5 A

16.2 Three resistors of 4 Ω, 6 Ω, and 8 Ω are connected in series to a 24 V battery. Find the total resistance and the current flowing through the circuit.

Given: R_1 = 4 Ω, R_2 = 6 Ω, R_3 = 8 Ω
V = 24 V

To find: R_s = ?, I = ?

Formula: $R_s = R_1 + R_2 + R_3$, I = V / R_s

Solution: R_s = 4 + 6 + 8
R_s = 18 Ω
I = 24 / 18
I = 1.33 A

Answer: R_s = 18 Ω, I = 1.33 A

16.3 Two resistors of 10 Ω and 30 Ω are connected in parallel across a 20 V power supply. Calculate the total resistance and the current drawn from the power supply.

Given: R_1 = 10 Ω, R_2 = 30 Ω
V = 20 V

To find: R_p = ?, I = ?

Formula: $1/R_p = 1/R_1 + 1/R_2$, I = V / R_p

Solution: $1/R_p = 1/10 + 1/30 = 3/30 + 1/30 = 4/30$
R_p = 30 / 4 = 7.5 Ω
I = 20 / 7.5
I = 2.67 A

Answer: R_p = 7.5 Ω, I = 2.67 A

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16.4 Four 1.5 V batteries are connected in series in a flashlight. Determine the total electromotive force (e.m.f.) supplied to the circuit.

Given: e.m.f. of one cell = 1.5 V
number of cells n = 4, joined in series

To find: E_{total} = ?

Formula: $E_{total} = n \times E$

Solution: $E_{total} = 4 \times 1.5$
$E_{total} = 6$ V

Answer: $E_{total} = 6$ V

16.5 A circuit has a 5 A current passing through a 7 Ω resistor. Calculate the power dissipated in the resistor and the energy consumed in 10 minutes.

Given: I = 5 A
R = 7 Ω
t = 10 min = 600 s

To find: P = ?, E = ?

Formula: $P = I^2R$, $E = P\,t$

Solution: $P = (5)^2 \times 7$
$P = 25 \times 7 = 175$ W
$E = 175 \times 600$
$E = 105\,000$ J
$E = 105$ kJ

Answer: P = 175 W, E = 105 kJ

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Formulae to Remember

- Current: $I = Q / t$; Ohm's law: $V = I\,R$
- Resistance: $R = \rho\, l / A$
- Series: $R_s = R_1 + R_2 + \ldots$; parallel: $1/R_p = 1/R_1 + 1/R_2 + \ldots$
- Power: $P = V\,I = I^2R = V^2/R$; energy: $E = P\,t$
- Cells in series: $E_{total} = n\,E$